

MST-Technologies and Processes

March, 25th, 2024

Duration: 120 min.

Please note

- write your name and matriculation number into the caption of every sheet
- use and write only on the front of the sheet
- use a separate sheet for every task
- write with a pen NOT with a pencil

Cheating

- any attempt of cheating results in immediate failure of the exam
- switch off your cell phones and store them in your backpack – a cell phone on the table will be treated as an attempt to cheating

Withdrawal or absence

- you may withdraw until the moment the tasks are revealed
- please see:
https://www.tf.uni-freiburg.de/en/studies-and-teaching/a-to-z-study-faq/de-registration-of-exams?set_language=en
- you find the relevant forms here:
<https://www.tf.uni-freiburg.de/de/studium-lehre/a-bis-z-studium/dokumente/WithdrawalfromExamsduetoIllnessMasterofScience.pdf>

Necessary Points: **55 of 110 in total**

Points: _____

Result: _____



1. Processes

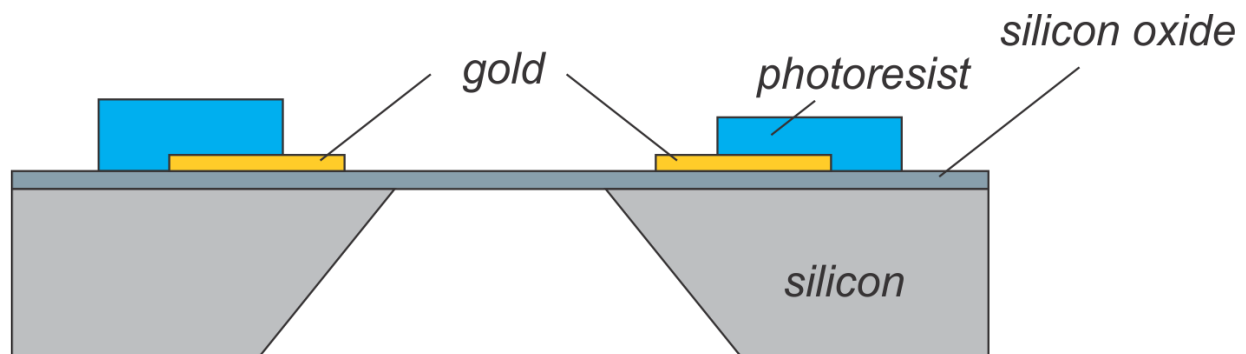
26

1.1 Manufacturing of a microfluidic sensor

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Sketch the manufacturing steps required to fabricate the following structure. This is cell capture device where cells are trapped within a microfluidic structure (defined by a photoresist) and assessed by electronic measurements using two electrodes (made of gold). The substrate on which the cells rest is silicon oxide (SiO_2) which was chosen due to its high biocompatibility. The setup is manufactured on a silicon wafer.

Sketch and briefly explain each step. Whenever you use a technique for the first time (e.g., photolithography), sketch it out in detail. If you use the same technique a second time, you do not have to sketch it out in detail a second time.



1.2 We have three different categories of processes in MST, name them and sort the following processes into the right category:

6

- Metallization
- Development during lithography
- Thermal oxidation



2. Cleanroom**22**

2.1 Draw a cross-section of a typical cleanroom. Label the relevant components. Make sure to include (at least) the following items:

7

- grey area
- white area
- plenum
- bypasses
- media supply

2.2 There are three types of cleanroom setups which are commonly used:

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- name them
- give one advantage for each
- give one disadvantage for each

2.3 Standard filter-fan units (FFUs) use four filtration effects. Name them and shortly explain each of them.

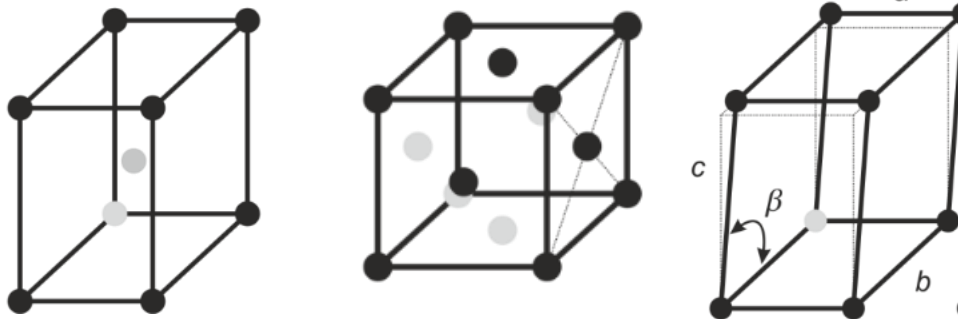
6



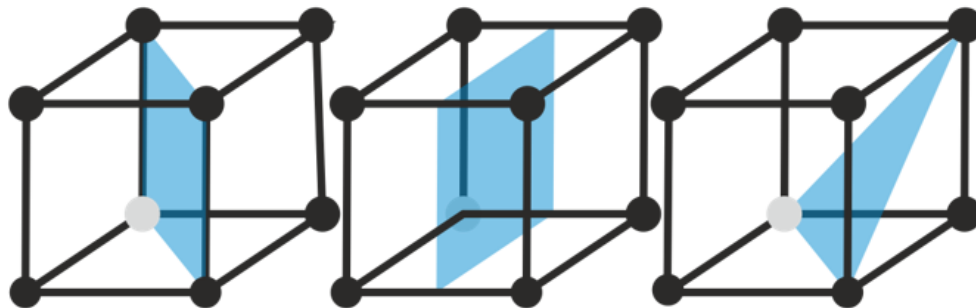
3. Semiconductor materials

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- 3.1 Name two common semiconductor materials 2
- 3.2 What is their most outstanding physical properties (for which they would be most likely chosen in an application) 2
- 3.3 Identify the following Bravais lattices (you can write your solution directly underneath the image) 6



- 3.4 Indicate the following crystal planes by their Miller indices (you can write your solution directly underneath the image) 6



- 3.5 Sketch and explain the float-zone process for manufacturing single-crystalline silicon. Make sure to label each component in your drawing. 8

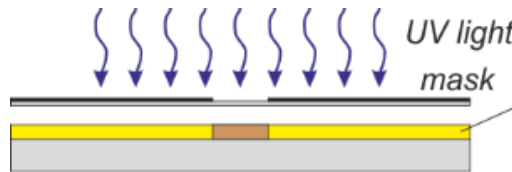


4. Photolithography

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- 4.1 The following image depicts two steps in a lithography workflow: the exposure (top) and the development step (below). Which type of photoresist was used here (what should the missing label read)?

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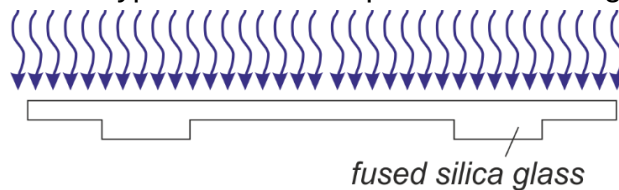


- 4.2 What would be a typical photoresist used for the setup depicted in 4.1?
- 4.3 What would you use as developer solution for the photoresist in the setup depicted in 4.1?
- 4.4 Which type of mask is depicted in this image?

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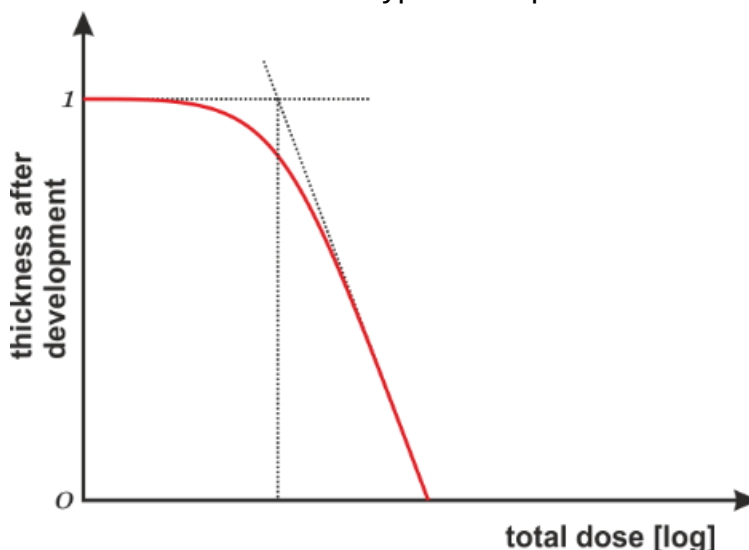


- 4.5 What are the two parameters that a mask can modulate on the incoming light?
- 4.6 Sketch the lithographic setup in projection lithography. Make sure to include all relevant labels in your sketch.
- 4.7 During characterization of an old bottle of photoresist via an exposure study, you obtain the following graph. Label the characteristic D_0 and D_{100} values and state the type of the photoresist.

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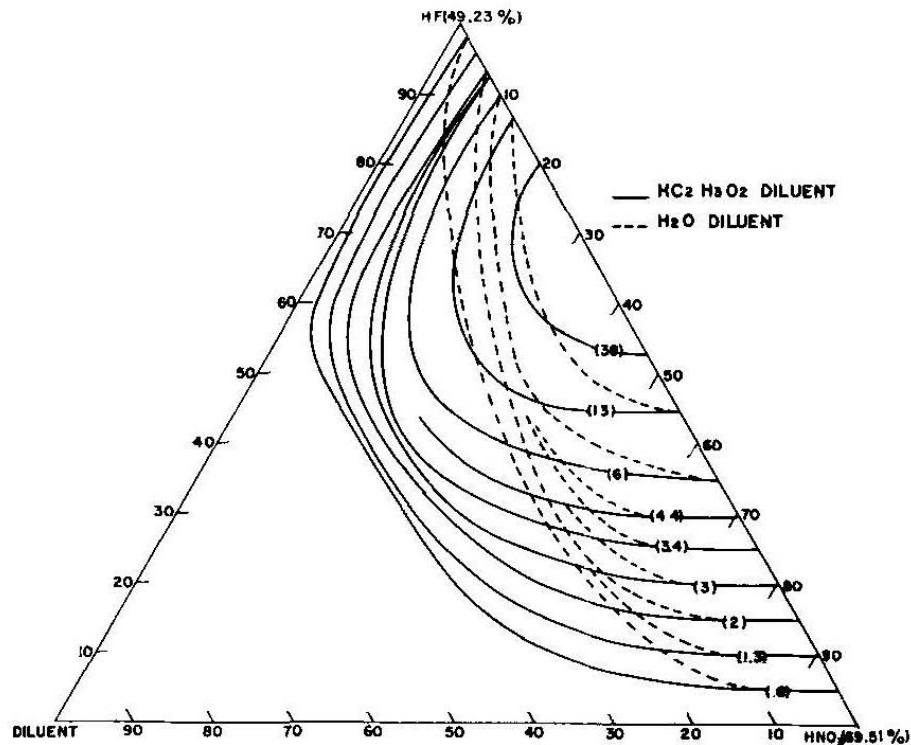
5. Etching

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- 5.1 Given the following etching diagram state three etching solutions which give an etching rate of 6 ($\mu\text{m}/\text{min}$) in acetic acid (solid lines) as diluent

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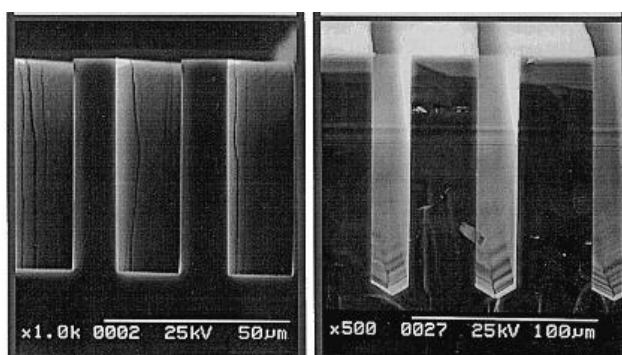


- 5.2 Name two typical etching alkaline etchants for anisotropic silicon etching.

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- 5.3 The following is an image of a (110) wafer having being etched by an alkaline etchant. The right image shows the (111) crystal planes at the bottom of the trench, the left one does not. Which etchant has been used in the left image?

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- 5.4 Sketch and describe the process steps of the Bosch process. Take care to also write down the actual etching reaction of silicon.

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